

The potential role of serotonergic mechanisms in the spinal oxytocin-induced antinociception



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ABSTRACT

The role of oxytocin (OXT) in pain modulation has been suggested. Indeed, hypothalamic paraventricular nuclei (PVN) electrical stimuli reduce the nociceptive neuronal activity (i.e., neuronal discharge associated with activation of Aδ- and C-fibers) of the spinal dorsal horn wide dynamic range (WDR) cells and nociceptive behavior. Furthermore, raphe magnus nuclei lesion reduces the PVN-induced antinociception, suggesting a functional interaction between the OXT and the serotonergic system. The present study investigated in Wistar rats the potential role of spinal serotonergic mechanisms in the OXT- and PVN-induced antinociception. In long-term secondary mechanical allodynia and hyperalgesia induced by formalin or extracellular unitary recordings of the WDR cells we evaluated the role of 5-hydroxytryptamine (5-HT) effect on the OXT-induced antinociception. All drugs were given intrathecally (i.t.). OXT (1×10^{-5} – 1×10^{-4} nmol) or 5-HT (1×10^{-3} – 1×10^{-1} nmol) prevented the formalin-induced sensitization, an effect mimicked by PVN stimulation. Moreover, administration of OXT (1×10^{-5} nmol) plus 5-HT (1×10^{-3} nmol) at ineffective doses, produced antinociception. This effect was antagonized by: (i) d(CH₂)₅[Tyr(Me)²,Thr⁴,Tyr-NH₂⁹]OVT (oxytocin receptor antagonist; 2×10^{-2} nmol); or (ii) methiothepin (a non-specific 5-HT_{1/2/5/6/7} receptor antagonist; 80 nmol). Similar results were obtained with PVN stimulation plus 5-HT (5×10^{-5} nmol). In WDR cell recordings, the PVN-induced antinociception was enhanced by i.t. 5-HT and partly blocked when the spinal cord was pre-treated with methiothepin (80 nmol). Taken together, these results suggest that serotonergic mechanisms at the spinal cord level are partly involved in the OXT-induced antinociception.

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1. Introduction

Oxytocin (OXT) has emerged as a mediator of endogenous analgesia and a potential drug to treat pain in humans (González-Hernández et al., 2014; Condés-Lara et al., 2016). Basic research shows that stimulation of the hypothalamic paraventricular nuclei (PVN) releases OXT at the spinal cord level and inhibits nociception (Condés-Lara et al., 2009; Condés-Lara et al., 2006; Eliava et al., 2016), an effect blocked by intrathecal (i.t.) administration of an oxytocin receptor (OTR)-pre-ferrering antagonist, d(CH₂)₅[Tyr(Me)²,Thr⁴,Tyr-NH₂⁹]OVT (OTA) (Breton et al., 2008; DeLaTorre et al., 2009). In agreement with these findings, electrical stimulation of the PVN or intrathecal (i.t.) OXT results in diminution of nociceptive behavior in several pain models (Eliava et al., 2016; Miranda-Cardenas et al., 2006; Pinto-Ribeiro et al., 2008).

Despite the relevance of OXT modulating nociception, few studies have investigated if OXT is able to inhibit indirectly the nociceptive input. For example, PVN electrical stimulation with the same parameters used to induce behavioral analgesia (Miranda-Cardenas et al., 2006) enhances c-Fos in the raphe magnus nuclei (RMg) (Condés-Lara et al., 2015), suggesting that a spinal serotonergic mechanism could be involved. Accordingly, lesion of the RMg reduces the spinal antinociception induced by PVN (Condés-Lara et al., 2012). Certainly, RMg activation is able to induce antinociception by directly inhibiting (via serotonergic fibers) the nociceptive input to the spinal dorsal horn, an effect mediated by 5-hydroxytryptamine receptors at this level (Hammond et al., 1985). These results point out a possible, spinal-level interaction between these two antinociceptive systems.

Therefore, the present study was designed to determine whether the antinociceptive effect of spinal OXT (endogenous or exogenous) could be mediated by serotonergic mechanisms. Since OXT is the main mediator involved in PVN-induced analgesia, we used this ligand and the selective OXT receptor antagonist, OTA. Furthermore, the endogenous ligand of the 5-HT receptors (5-HT or serotonin) or the non-specific 5-HT_{1/2/5/6/7} receptor antagonist, methiothepin, was used.

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The experiments were carried out using the: (i) long-term secondary mechanical allodynia and hyperalgesia induced by formalin; and (ii) extracellular unitary recordings of the spinal dorsal horn (SDH) wide dynamic range (WDR) cells. In both cases, some animals were implanted (in the PVN) with a stimulating electrode in order to evaluate the role of serotonergic mechanisms in the SDH.

2. Materials and methods

2.1. Animals

Male Wistar rats (280–310 g) were used. The animals were housed at 22 ± 2 °C, with food and water *ad libitum*. Our Institutional Ethics Committee approved all animal procedures, and we followed the ethical guidelines for the care and use of laboratory animals by the IASP (Zimmermann, 1983) and the NIH (86-23). Efforts were made to minimize the number of animals used and their suffering.

2.2. Behavioral experiments

2.2.1. Intrathecal surgery and electrode implantation in the PVN

In order to insert an intrathecal (i.t.) catheter for drug administration, the animals were anesthetized with a ketamine/xylazine mixture (70/6 mg/kg, i.p.) and placed in a stereotaxic head holder to maintain them in a vertical position. The atlanto-occipital membrane was exposed (LoPachin et al., 1981). The membrane was pierced, and a PE10 catheter (8.5 cm long) was passed intrathecally to reach the lumbar L2–L4 segments; then the wound was sutured, and the catheter was fixed to the occipital bone with acrylic dental cement. Animals were allowed to recover for 6 days. Only animals showing symmetric effects of i.t. xylocaine (10 μ l at 2%) were used. All the i.t. administrations were composed of the initial injection of 10 μ l of the drug followed by 10 μ l of saline solution, which remained in the catheter for further administrations.

Six days after i.t. surgery, the animals were anesthetized with ketamine and xylazine (70/6 mg/kg, i.p.), and a bipolar electrode was stereotactically implanted into the left PVN (7.0 mm AP from interaural, 0.2 mm lateral, and 2 mm height) (Paxinos and Watson, 1998). The electrodes and connectors were fixed to the skull by means of acrylic dental cement. Animals were allowed to recover from surgery for 6 days before formalin injection. The PVN stimulation used to induce antinociception began 6 days after formalin injection (see Section 2.2.2). This stimulation consisted of a train of pulses of 1-ms duration at 60 Hz during a 6-s interval. The stimulation intensity (100–300 μ A) was adjusted for each rat, depending on the behavioral responses and taking care to avoid side effects.

2.2.2. Induction of long-term secondary mechanical allodynia and hyperalgesia

Von Frey filaments were used to assess the long-term secondary mechanical allodynia and hyperalgesia as described by Jolivald et al. (2006) and Ambriz-Tututi et al. (2009). Briefly, long-term secondary mechanical allodynia and hyperalgesia were induced by subcutaneous (s.c.) injection (30 G) of 50 μ l formalin (1%) into the dorsal surface of the right hind paw, and behavioral tests were performed 1, 3, 6, 9, 12, and 15 days later.

The withdrawal responses to Von Frey filaments (Stoelting Co, Wood Dale, IL, USA) were measured before and after formalin injection. In all cases, the animals were habituated for at least 30 min over 3 days. The rats were placed in testing cages with a wire-mesh bottom, and baseline measurements were recorded. Two Von Frey filaments (bending forces of 10 and 250 mN) were applied 10 times in each testing set at the base of the third toe on the plantar surface of both paws. Three trials were completed and averaged to determine the number of paw withdrawal responses. Allodynia and hyperalgesia were considered secondary, as stimuli with the von Frey filaments were applied to sites different

from the formalin injection. Under normal conditions, a force of 10 mN does not activate cutaneous nociceptors (Leem et al., 1993) and, therefore, it does not cause paw withdrawal in normal animals. Accordingly, the occurrence of responses in the rats indicates allodynia. On the other hand, a force of 250 mN is considered a noxious stimulus, and hyperalgesia occurs when the number of withdrawals is increased. In all cases, the baseline responses to von Frey filaments were recorded before i.t. surgery; also, the nociceptive behavior was evaluated before and 6 days after electrode implantation.

2.2.3. Study design

The nociceptive behavior induced by von Frey Filaments was evaluated in both paws (ipsilateral and contralateral to formalin injection) before and 10 min after i.t. drug administration or PVN stimulation. To exclude the possibility that the effect of i.t. drugs on paw withdrawal responses was influenced by the vehicle, a group of animals ($n = 6$) was injected with saline. In all cases, the experiments with spinal administration of 5-HT, OXT, methiothepin and OTA or PVN stimulation started on the sixth day after formalin injection, since the nociceptive behaviors are well established at this time (Fu et al., 2000; Ambriz-Tututi et al., 2011).

To assess if the i.t. injection of 5-HT ($n = 8$) or OXT ($n = 8$) can modify the occurrence of formalin-induced secondary mechanical allodynia and hyperalgesia, animals received an i.t. administration of vehicle or increasing doses of 5-HT (0.001–0.1 nmol) or OXT (1×10^{-5} – 1×10^{-4} nmol) 10 min before the paw withdrawal evaluation. Each drug dose treatment was separated by a 72 h washout period.

In another group ($n = 6$) we analyzed the effect of i.t. administration of ineffective doses of: (i) OXT (1×10^{-5} nmol); (ii) 5-HT (0.001 nmol); or (iii) OXT (1×10^{-5} nmol) *plus* 5-HT (0.001 nmol). We were also interested in determining whether the anti-allodynic effect produced by the co-administration of 5-HT *plus* OXT could be blocked by pre-treatment with 2×10^{-2} nmol OTA (Miranda-Cardenas et al., 2006) or 80 nmol methiothepin (Lozano-Cuenca et al., 2005). In this case, the animals received the i.t. OTA or methiothepin 10 min before the i.t. 5-HT *plus* OXT. Each drug dose treatment was separated by a 72 h washout period.

To determine if electrical stimulation of the PVN can modify the occurrence of formalin-induced secondary mechanical allodynia, a group of rats ($n = 8$) received an electrical stimulation of the PVN 15 min before the behavioral test. Three days later, these animals received 5-HT (1×10^{-3} nmol, i.t.), and 15 min later they were tested for secondary mechanical allodynia. Three days later, i.t. 5-HT (1×10^{-3} nmol) *plus* PVN stimulation was given, and the behavioral test was performed 15 min later. Finally, 3 days later, these animals received methiothepin (80 nmol; i.t.) 10 min before the i.t. 5-HT *plus* PVN stimulation and the behavioral test was performed again 15 min later.

It is important to point out that we analyzed different treatments in the same animal. Certainly, a previous report showed that 48 h is sufficient to wash out the potential residual effect of OXT (Miranda-Cardenas et al., 2006; Hicks et al., 2014). But, in order to exclude any residual effect due to the previous treatment, we measured the basal response one day before each drug test.

2.3. Electrophysiological experiments

2.3.1. Surgical procedures

Animals were anesthetized with 8% (v/v) sevoflurane delivered in a 3:1 ratio of N₂O and O₂. An intratracheal cannula was inserted for artificial ventilation and to maintain the anesthesia throughout the experiments. Rats were mounted on a stereotaxic frame. The spinal cord was fixed at the lumbar vertebrae to perform a laminectomy at spinal cord segments L2–L4. The dura was removed. The subsequent protocols were performed under 2.5% sevoflurane. End tidal CO₂ was monitored (Capstar-100, CWI Inc., Ardmore, PA, USA), and core body temperature was maintained at 38 °C using a circulating water pad.

2.3.2. Extracellular unitary recordings and electrical PVN stimulation

Concentric stainless steel stimulation electrodes (1 M Ω) were placed in the PVN at 7.0 mm AP from interaural, 0.2 mm lateral, and 2 mm height (Paxinos and Watson, 1998). Moreover, glass microelectrodes (4–10 M Ω ; 2% Pontamine Sky Blue; 1 M KCl) were lowered (200–1000 μ m from the surface) into the superficial left laminae of the SDH to search for single-unit discharges evoked by gentle mechanical stimulation. For each recorded cell, the specific somatic receptive field (RF) was located by tapping on the entire ipsilateral hind paw and toes. Once the RF was located, we inserted two needles (27 G) attached to a stimulus isolator unit to deliver electrical stimulation (Grass S11 Stimulator). The stimuli consisted of 1-ms electrical pulses at 0.5 Hz with increasing intensities to determine the threshold for activating A δ - and C-fibers. All cells studied were classified as WDR cells.

The extracellular neuronal activity was amplified, digitalized, and discriminated using CED hardware and Spike 2 software (Cambridge Electronic Design). The activity of the recorded cells was analyzed with post-stimulus time histograms to identify action potentials with response latencies related to activation of A β - (0–20 ms), A δ - (21–90 ms) and C-fibers (91–300 ms). We proceeded to analyze the effect of the PVN stimulation on the evoked responses induced by RF stimulation. The number of action potentials that occurred in response to 20 RF stimuli was compared before and after PVN stimulation. Fig. 4 shows an example of the experimental procedure. The PVN stimuli were delivered over a 6-s interval at 60 Hz with a pulse duration of 1 ms and 300 μ A.

2.3.3. Experimental protocol to test the effects of spinal administration of serotonin on the PVN-induced inhibition of nociceptive responses of WDR neurons

To test the effect of the spinal serotonergic system on the antinociception by PVN, we first analyzed the effect of spinal 5-HT administration (10 nmol, $n = 7$; 25 nmol, $n = 5$ and 50 nmol, $n = 5$) on the evoked nociceptive activity of the WDR neurons (20 stimuli delivered in the RF) over a 40-min interval: at 0, 5, 10, 15, 20, 30 and 40 min after 5-HT administration. In addition, the effect of the spinal 5-HT (10 nmol, $n = 7$; 25 nmol, $n = 5$ and 50 nmol, $n = 5$) on the PVN-induced antinociception was evaluated for 40 min at 0, 5, 10, 15, 20, 30 and 40 min after 5-HT. To exclude any effect due to time, a control curve was made ($n = 5$).

2.3.4. Experimental protocols to test the effects of spinal methiothepin on the PVN-induced inhibition of nociceptive responses of WDR cells

To test for a potential role of spinal 5-HT receptors in the PVN-induced antinociception, we analyzed the effect of spinal methiothepin administration (80 nmol; $n = 3$) on the PVN-induced antinociception for 40 min: at 0, 5, 10, 15, 20, 30 and 40 min after methiothepin administration. The effect of spinal methiothepin *per se* on the evoked WDR neuronal activity was evaluated ($n = 5$).

2.4. Histological reconstruction of recording sites

The position of the cells recorded was marked by iontophoretic injection of Pontamine Sky Blue (cathodic current 10–15 μ A, 30 min). The animals received an overdose of pentobarbital (80 mg/kg) and were perfused transcardially with saline solution followed by 10% formaldehyde (~200 ml of each). Brains were postfixed in 10% formaldehyde and cut in 40- μ m serial coronal sections with a freezing microtome (Leica Instruments BmbH, Nussloch, Germany). Recording sites were observed with a light microscope.

2.5. Compounds

Apart from the anesthetics, the drugs used in the present study were: OXT (syntocinol®, 5 IU/ml, Novartis); d(CH₂)₅[Tyr(Me)²,Thr⁴,Tyr-NH₂⁵]OVT (gift from Prof. Maurice Manning); 5-hydroxytryptamine

hydrochloride and methiothepin mesylate (Sigma Chemical Co., St. Louis, MO, USA). All compounds were dissolved in saline.

2.6. Data presentation and statistical evaluation

All data are presented as the mean $\pm$ standard error of the mean (SEM). In all behavioral and electrophysiological experiments, statistical significance was accepted at $p < 0.05$. In the behavioral experiments, the mean response to von Frey filaments was obtained from the ipsilateral and contralateral paws. In this case, a one-way ANOVA followed (if applicable) by the Bonferroni test was performed.

In the electrophysiological experiments, the RF-evoked activities were quantified by classifying the action potentials according to their response latency. The changes in neuronal activity are expressed as percent change from the basal response; the differences were compared with a two-way ANOVA followed (if applicable) by the Bonferroni test. In order to obtain an integrated measurement of the effect induced by 5-HT or 5-HT *plus* PVN stimuli in the electrophysiological experiments, we calculated the area under curve (AUC) using the trapezoidal rule; the differences were compared with a one-way ANOVA. Furthermore, in the case of the effect of methiothepin on the PVN-induced antinociception a Friedman test was used.

3. Results

3.1. Behavioral experiments

3.1.1. Formalin-induced secondary allodynia and hyperalgesia

Subcutaneous (s.c.) injection of 1% formalin, but not saline, into the dorsal surface of the right hind paw produced secondary mechanical allodynia and hyperalgesia (Fig. 1), not only in the injected but also in the contralateral paw. This was seen as a bilateral increase in paw withdrawal responses to 10 and 250 mN von Frey filaments, which was significant one day after formalin injection and lasted for at least 15 days ($p \leq 0.05$). The paw withdrawal responses were similar in both paws. Additionally, the secondary mechanical allodynia in the rats implanted with a PVN electrode (Fig. 1C) was similar to nociceptive responses induced in the animal that only received the formalin injection.

3.1.2. Effect of 5-HT and OXT on formalin-induced secondary allodynia and hyperalgesia

Intrathecal pre-treatment on days 6, 9 and 12 with increasing doses of OXT (1×10^{-5} – 1×10^{-4} nmol) significantly ($p \leq 0.05$) reversed 1% formalin-induced secondary mechanical allodynia (Fig. 2A) and hyperalgesia (Fig. 2B) in both paws. Similar results ($p \leq 0.05$) were obtained with increasing doses (on days 6, 9, 12, 15) of 5-HT (0.001–0.1 nmol) (Fig. 2C and D). Since OXT and 5-HT inhibited the sensitization induced by formalin, the role of oxytocin receptors (OTR) and 5-HT receptors was evaluated. In the following experiments with PVN electrode implantation we only evaluated allodynia because the induction of hyperalgesia by formalin was unreliable, an effect probably related to the several surgeries performed on these animal groups.

3.1.3. Effect of methiothepin and OTA on the OXT plus 5-HT-induced antiallodynic effects

Intrathecal pre-treatment with the 5-HT (1×10^{-3} nmol) *plus* OT (1×10^{-5} nmol) at doses that do not, alone, have any effect on the formalin-induced sensitization was able to block the allodynia in both paws ($p \leq 0.05$, Fig. 3A). This effect was blocked by i.t. OTA (2×10^{-2} nmol) or methiothepin (80 nmol). We need to keep in mind that at the dose used, neither antagonist by itself has any effect on the formalin-induced sensitization.

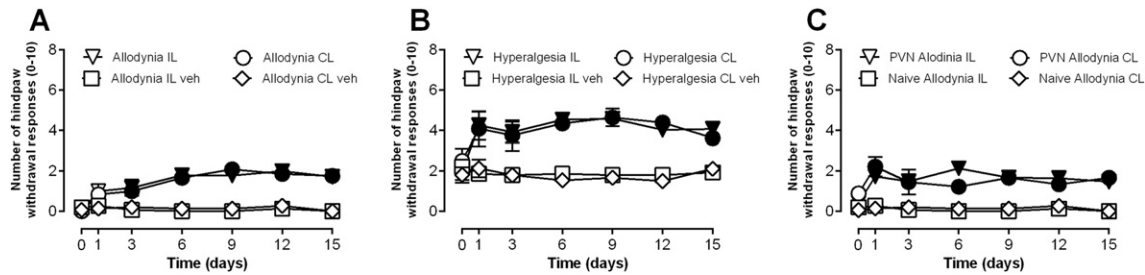


Fig. 1. Formalin-induced sensitization of the ipsilateral and contralateral paw. Time course of the mean number of paw withdrawal responses on the ipsilateral (IL) and contralateral (CL) side following applications of Von Frey filaments of two different bending forces, 10 mN (allodynia, panel A) and 250 mN (hyperalgesia, panel B), after subcutaneous administration of 1% formalin ($n = 6$). In panel C we show the effect of PVN electrode implantation ($n = 6$) on the allodynia induced by 1% formalin. For clarity, black symbols instead of white symbols represent significantly different responses ($p < 0.05$) vs. control (day 0; basal response before formalin).

3.1.4. Effect of methiothepin on the antiallodynic effect induced by PVN electrical stimulation plus 5-HT

As shown in Fig. 3B, PVN stimulation decreases the number of hindpaw withdrawal responses. A similar result was obtained with 1×10^{-3} nmol 5-HT. Furthermore, i.t. pre-treatment with 1×10^{-3} nmol 5-HT plus PVN stimulation not only blocked the allodynia produced by formalin but also potentiated the effect of PVN alone. Interestingly, when we evaluated the role of 5-HT receptors, we found that i.t. methiothepin (80 nmol/rat) abolished the antinociception induced by 5-HT and PVN stimuli ($p \leq 0.0001$).

3.2. Electrophysiological experiments

3.2.1. Electrical PVN stimulation

A total of 47 cells (one per rat) located at the L4–L5 level of the spinal cord dorsal horn were tested. These neurons were characterized

according to the properties of their responses to peripheral receptive field (RF) stimulation and were classified as WDR neurons since they presented A β -, A δ -, and C-fiber responses (Fig. 4). As previously reported (Condés-Lara et al., 2006), electrical PVN stimulation selectively reduces only the nociceptive (A δ - and C-fiber) activity evoked by peripheral RF stimulation.

3.2.2. Effects of spinal administration of 5-HT or 5-HT plus PVN stimulation on the spinal cord unit activity

In animals pretreated with 5-HT (10–50 nmol) or 5-HT plus PVN stimuli, the neuronal activity of A β -fibers remained unaltered (Fig. 5A and B). Furthermore, although 5-HT at 25 and 50 nmol was able to reduce the neuronal activity of A δ -fibers (Fig. 5C), a transitory increase was observed at min 5 with the 10 and 25 nmol doses. In addition, when PVN stimulation was given in the animals pretreated with 10–25 nmol 5-HT (Fig. 5D), a decrease in the neuronal activity was

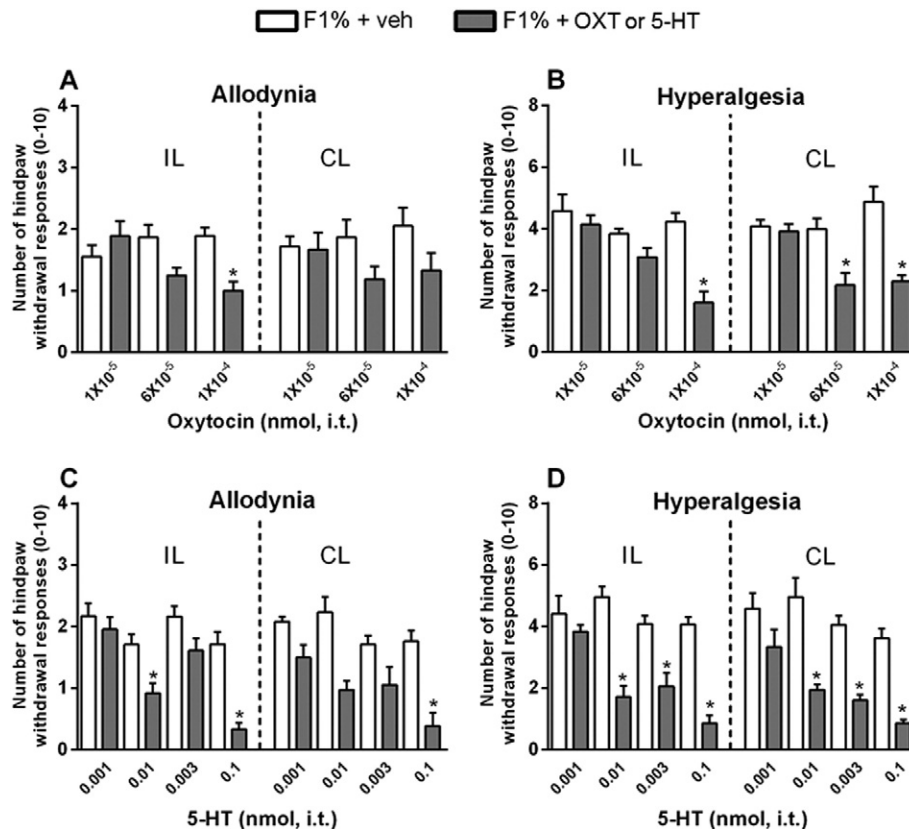


Fig. 2. Intrathecal oxytocin or 5-HT inhibits the sensitization induced by formalin. Effect of treatment (starting from day 6, 9, 12, 15 after formalin injection) with oxytocin (A and B) (OXT, 1×10^{-5} – 1×10^{-4} nmol, $n = 8$) or 5-HT (C and D) (0.001–0.1 nmol, $n = 8$) on 1% formalin (F1%)-induced secondary mechanical allodynia and hyperalgesia in rats. Note that the OXT was able to revert the allodynia (A) but not the hyperalgesia (B). Similar results were obtained with 5-HT (C and D). Data are expressed as the ipsilateral (IL) and contralateral (CL) paw withdrawal responses to the von Frey filament examination (10 and 250 mN) of the plantar surface of rat paws before (veh, vehicle) and after OXT (A and B) or 5-HT (C and D). * $p \leq 0.05$ vs. F1% + veh.

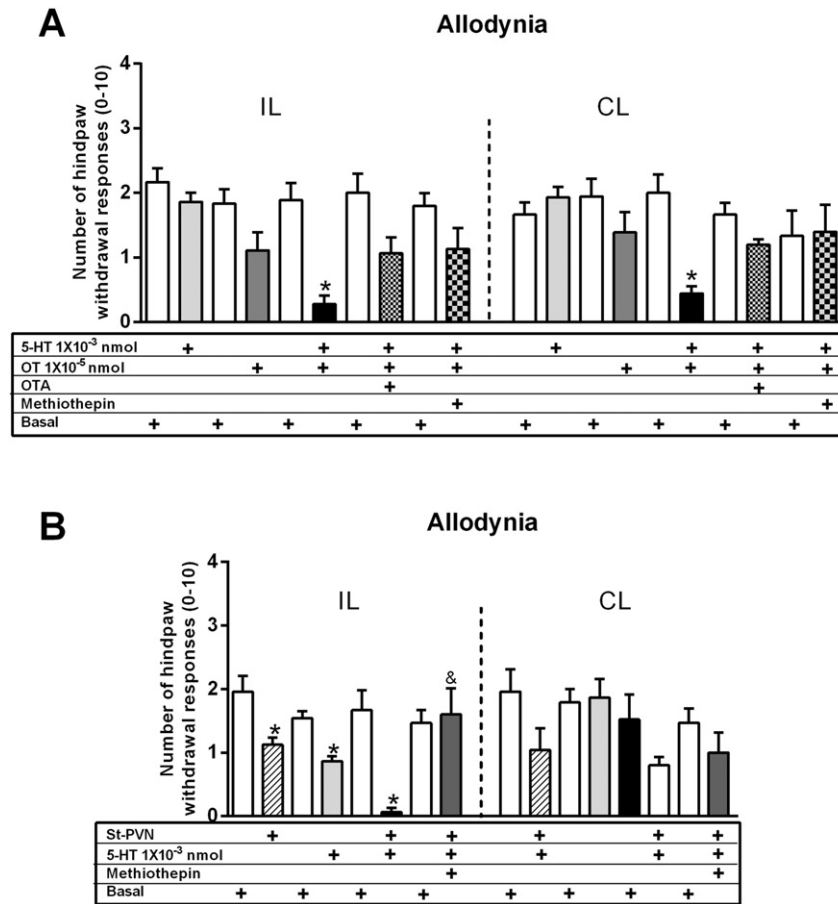


Fig. 3. The role of oxytocin receptors or 5-HT receptors in the antinociception induced by oxytocin, 5-HT or PVN stimulation. Effect of $d(CH_2)_5[Tyr(Me)^2, Thr^4, Tyr-NH_2^5]OVT$ (OTA 2×10^{-2} nmol, day 12) and/or methiothepin (MET 80 nmol, day 15) on the anti-allodynic effects induced by: (A) oxytocin (1×10^{-5} nmol), (B) PVN stimulation (St-PVN) and (A and B) 5-HT (1×10^{-3} nmol) administration. In (A) OTA was given at day 12 and MET was given at day 15 ($n = 6$). In (B) St-PVN was delivered at day 12 and MET at day 15 ($n = 8$). Data are expressed as the ipsilateral (IL) and contralateral (CL) paw withdrawal responses to the applications of von Frey filaments (10–250 mN) on the plantar surface of rat paws after formalin injection. * $p \leq 0.05$ vs. basal group, & $p \leq 0.05$ vs. St-PVN + 5-HT. St-PVN, electrical stimulation of the PVN.

achieved, but a transitory increase of the neuronal activity was observed with 50 nmol 5-HT. In the case of C-fibers (Fig. 5E and F) spinal 5-HT or 5-HT plus PVN stimulation diminished the neuronal activity induced by RF stimulation.

Since we observed a biphasic effect induced by 5-HT in the A δ -fibers activity we decided to calculate the area under curve (AUC) in two periods; the first AUC corresponding between basal and 10 min and the second AUC between 20 and 40 min (Fig. 5G and 5H). Certainly, the AUC is a more accurate indicator of the global effect of a treatment (in our case is an index of antinociception or pronociception induced by 5-HT or 5-HT plus PVN stimulation). When we analyzed the activity of A δ -fibers, 5-HT alone had no effect on the evoked nociceptive activity during the first 10 min. However, between 10 and 40 min, 25 or 50 nmol 5-HT was able to significantly reduce the neuronal firing associated with activation of A δ -fibers. It is interesting to note that 50 nmol 5-HT abolished the PVN-induced blockade of A δ -fiber activity (Fig. 5G; right panel). In the case of C-fibers (Fig. 5H), during the first 10 min spinal 5-HT has no effect, but C-fiber diminished between 10 and 40 min with 25 and 50 nmol 5-HT. The combination of 5-HT plus PVN stimuli reduced the C-fiber activity during the first 10 min as well as between 10 and 40 min. However, the AUC obtained from the 10- to 40-min time period suggests that the neuronal C-fiber blockade induced by PVN stimuli was enhanced with 50 nmol 5-HT.

3.2.3. Effects of spinal methiothepin on the PVN-induced antinociception

Fig. 6 shows that after methiothepin (80 nmol) administration ($n = 3$), the antinociception induced by PVN stimulation was partially

reversed. Indeed, spinal methiothepin did not alter the activity evoked by A β -fibers before and after the PVN stimuli (Fig. 6). On the other hand, the inhibition by PVN stimulation of the neuronal activity of A δ and C-fibers was partially reverted and readily observed by 5 min after methiothepin administration. It is important to point out that spinal administration of methiothepin ($n = 5$) at the dose used (80 nmol) did not have any significant effect on the peripheral evoked WDR neuronal activity (Fig. 7).

4. Discussion

4.1. General

Our study shows that the long-term, secondary mechanical allodynia and hyperalgesia induced by formalin (Fig. 1) are inhibited by spinal administration of OXT and 5-HT (Fig. 2), suggesting a potential role of oxytocinergic and serotonergic receptor-dependent mechanisms (Fig. 3). Accordingly, in the electrophysiological recordings of WDR cells, the nociceptive responses of these second-order neurons (i.e., neuronal discharges of A δ - and C-fibers) are strongly suppressed when PVN-electrical stimuli are applied with 5-HT (Fig. 5). Since in the behavioral experiments, the OXT- (Fig. 3A) and most importantly, the PVN-induced (Fig. 3B) antinociception was moderately blocked not only by spinal OTA but also by methiothepin (a 5-HT non-selective receptor antagonist), an interaction between these two antinociceptive systems is suggested. A hypothesis reinforced by the fact that in the electrophysiological experiments, the PVN-induced blockade of the

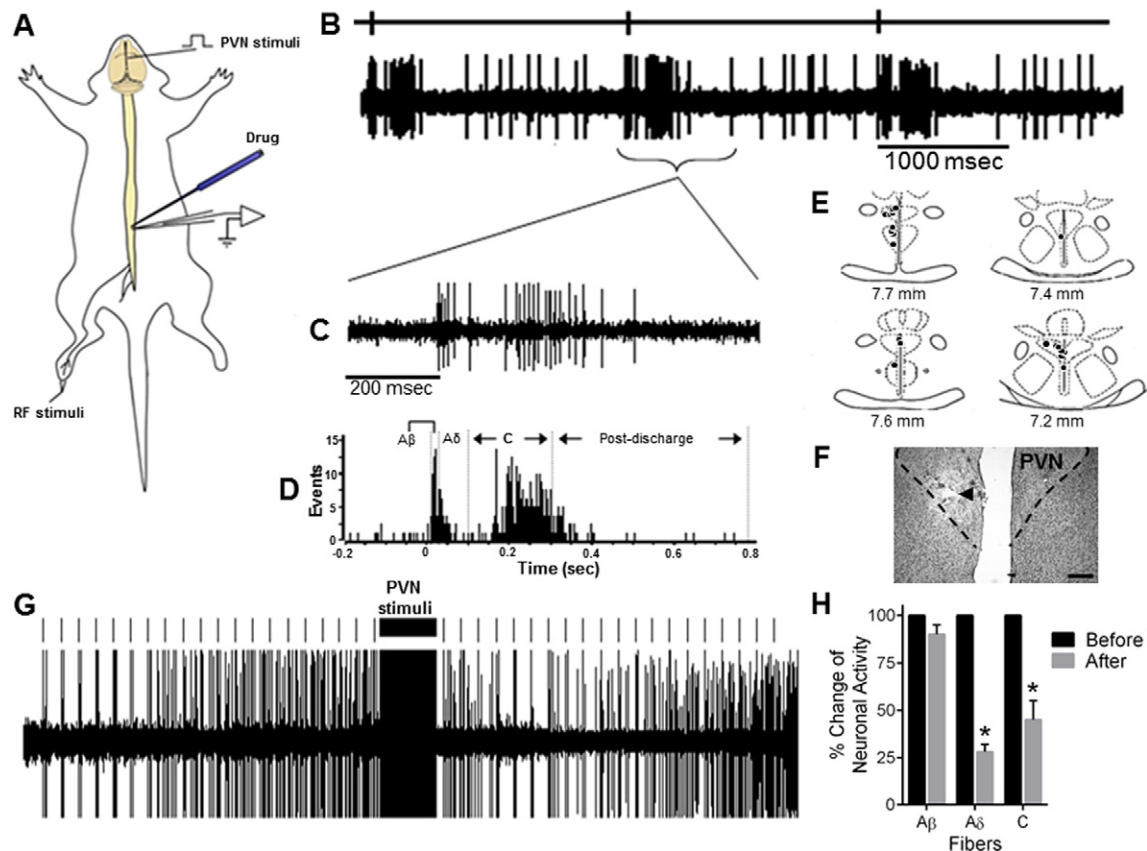


Fig. 4. Experimental set-up used to record WDR neurons. Neural responses to receptive field electrical stimulation (RF-st). (A) Representation of the experimental setup. Spinal dorsal horn wide dynamic range neurons (WDR) were recorded using a glass micropipette (4–8 MΩ) inserted into the spinal dorsal horn ipsilateral to the PVN electrode. (B) This panel shows three stimulus artifacts (above) induced by RF-st and the neuronal responses evoked (below) by the RF stimulation of an identified WDR neuron. (C) A single response induced by one electrical stimulus on the RF. (D) The peri-stimulus time histogram obtained from 20 responses to 20 RF-st; the time latencies depict the different fiber components (Aβ-fiber, 5–20 ms; Aδ-fibers, 21–90 ms; and C-fibers, 91–350 ms). (E) The location of the stimulated points in the PVN is shown. (F) Microphotography of the PVN showing the lesion (◄) induced by the stimulating electrode. (G) Typical electrophysiological recording obtained showing the WDR cell responses to 20 RF-st before and 20 RF-st after PVN electrical stimulation; note that the neuronal evoked activity after PVN stimulation is reduced. (H) This PVN-induced inhibition was selective for the neuronal activity related to the activation of Aδ- and C-fibers (nociceptive fibers) but not of Aβ-fibers (non-nociceptive fibers) ($n = 5$). The PVN stimuli were delivered over 6 s at 60 Hz and 300 μ A with a pulse duration of 1 ms. These parameters allowed us to observe: (i) an inhibitory action on the nociceptive component (Aδ- and C-fibers) of the WDR neurons; and (ii) a recovery of the cell responses < 1 min after PVN stimulation.

neuronal activity of Aδ- and C-fibers was partially reverted by methiothepin (Fig. 6).

These findings shed further light on the mechanism involved in the antinociception induced by OXT, and they represent the first *in vivo* evidence suggesting the existence, at the spinal cord level, of an interaction between the oxytocinergic and serotonergic systems that modulates nociception.

4.2. Spinal OXT and 5-HT inhibited the formalin-induced long-term sensitization

As previously reported (Ambriz-Tututi et al., 2009; Ambriz-Tututi et al., 2011), formalin injection at 1% produces acute (1 h) nociceptive behavior (not shown here), as well as secondary mechanical allodynia (3–15 days) and long-term (1–15 days) hyperalgesia in both paws (Fig. 1). Since the long-term allodynia and hyperalgesia induced by formalin was partially reverted by OXT (Fig. 2A and B) or 5-HT (Fig. 2C and D), the simplest interpretation of this finding may suggest that OXT and 5-HT are acting at the spinal level via their respective receptors. Indeed, spinal oxytocin receptors (OTR) are considered the main mediator involved in the PVN-induced antinociception (González-Hernández et al., 2014; Condés-Lara et al., 2006; Miranda-Cardenas et al., 2006). Nevertheless, several studies have suggested that other neuromediators could be contributing to this effect (Condés-Lara et al., 2012; Ge et al., 2002; Rojas-Piloni et al., 2007; Yang et al., 2007).

In this regard, electrophysiological experiments on the WDR neurons have demonstrated that lesion of the raphe magnus nuclei (RMg) (a main source of endogenous 5-HT) diminishes the PVN-induced antinociception (Condés-Lara et al., 2012). These results implied that PVN electrical stimulation blocks the nociceptive responses of the WDR cells not only directly but also indirectly by activating the RMg, a thesis reinforced by the fact that PVN stimuli are able to enhance c-Fos expression in the RMg (Condés-Lara et al., 2015). Together, these studies point to a potential role for serotonergic descending mechanisms in the PVN-induced antinociception. Of course, activation of spinal 5-HT receptors in the long-term secondary allodynia and hyperalgesia induced by formalin has been extensively studied (Bravo-Hernández et al., 2012; Cervantes-Durán et al., 2013; Godínez-Chaparro et al., 2011; Godínez-Chaparro et al., 2013), but no one has shown that these two systems could be interacting.

4.3. Possible involvement of spinal OTR and 5-HT receptors in the long-term sensitization induced by formalin

Co-administration of OXT (1×10^{-5} nmol) plus 5-HT (1×10^{-3} nmol) (Fig. 3A) produced an anti-allodynic effect not observed when either OXT or 5-HT was given alone at these concentrations, suggesting a role for OTR and 5-HT receptors. In fact, pre-treatment with OTA or methiothepin prevented this effect (Fig. 3A), supporting not only the role of these receptors but also a potential interaction between the oxytocinergic and serotonergic system. Certainly, we could argue

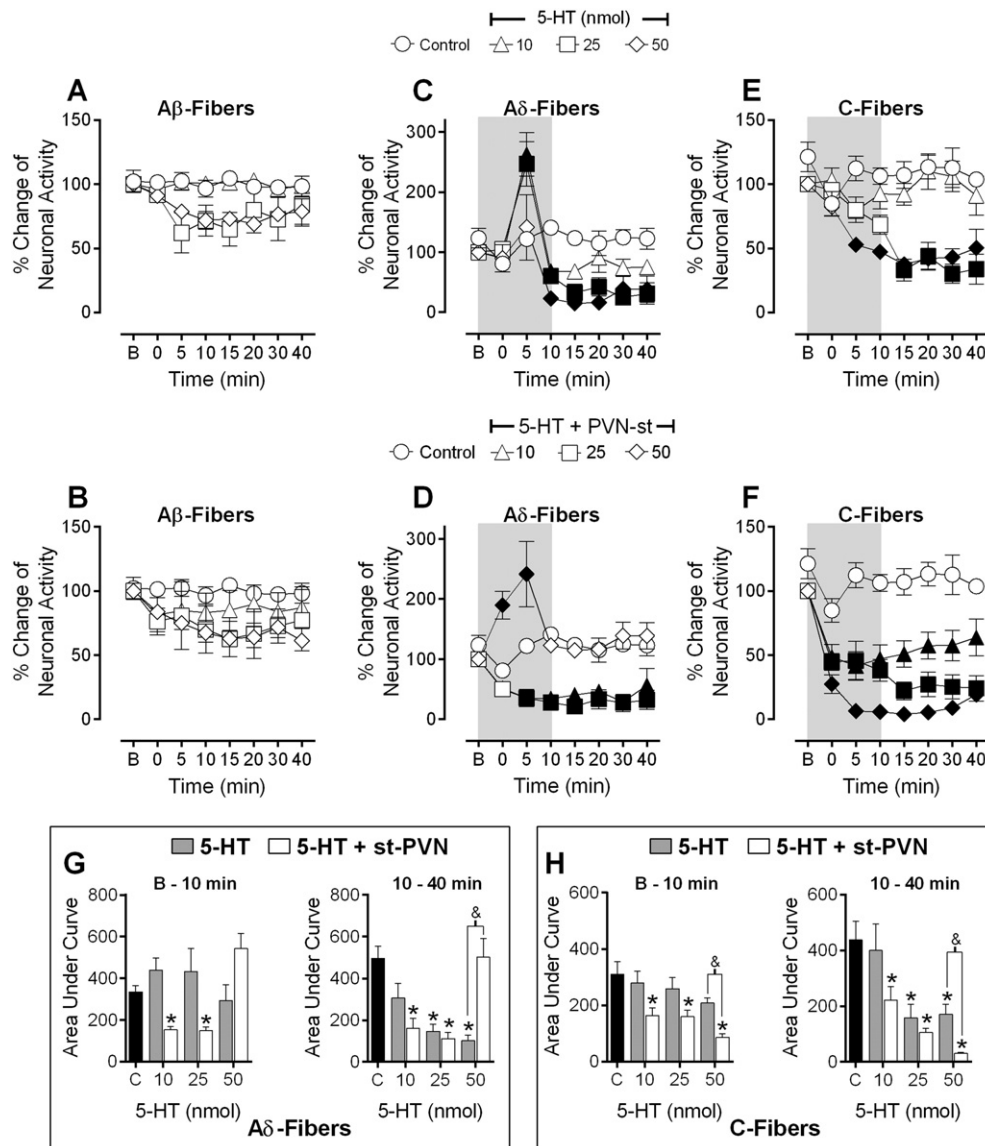


Fig. 5. Spinal 5-HT enhances the antinociception induced by PVN stimulation. Percent change in the neuronal evoked activities after modulation of nociceptive responses in WDR neurons by administration of 5-HT (10 nmol, $n = 7$; 25 nmol, $n = 5$ and 50 nmol, $n = 5$) or 5-HT plus PVN stimulation (st-PVN) (at the same 5-HT doses and n). The graphs show the time course of the % change of neuronal activity of A β -, A δ - and C-fibers after administration of nothing (control, $n = 5$) or 5-HT (A, B and C) or 5-HT + PVN stimulation (D, E and F). Furthermore, panels G and H show that when PVN stimuli are applied in rats pretreated with 5-HT, the antinociception is enhanced. For clarity, empty symbols in panels A to F depict either control responses (white circle) or non-significant responses (white, triangle, square and rhombus) vs. control responses. Black symbols (triangle, square and rhombus) represent significantly different responses ($p < 0.05$) vs. control responses (white circle). * $p < 0.05$ vs. C (control); $\&p < 0.05$ vs. 5-HT treatment.

that this effect is due to a synergistic effect rather than an interaction between the oxytocinergic and serotonergic mechanisms.

On the other hand, since the major source of OXT is the PVN, it was important to demonstrate that stimulation of the PVN was able to emulate the behavioral antinociception produced by exogenous OXT. Consistent with this, electrical stimulation of the PVN induced behavioral antinociception (Fig. 3B) (Condés-Lara et al., 2012; Shiraishi et al., 1995; Yirmiya et al., 1990); as previously reported (Condés-Lara et al., 2009; Condés-Lara et al., 2006; Eliava et al., 2016; Breton et al., 2008; DeLaTorre et al., 2009; Miranda-Cardenas et al., 2006), OXT is the neuromediator involved. Furthermore, our data (Fig. 3B) showed that PVN stimuli plus 5-HT produced an antiallodynic effect of the same magnitude as exogenous OXT plus 5-HT. Since this effect was partially reverted by methiothepin, the role of 5-HT receptors is suggested, a hypothesis supported by the results obtained in the electrophysiological experiments (see Discussion Section 4.4).

One important issue about the use of methiothepin as a 5-HT receptor antagonist is the fact that this compound has affinity not only

for 5-HT_{1/2/5/6/7} receptors but also for α_1 -adrenoceptors (pKi: 9.3) and α_2 -adrenoceptors (pKi: 7.3) (Leysen, 1985). However, under the experimental conditions used in our behavioral experiments (Fig. 3) and since we used serotonin (*i.e.*, the endogenous ligand), we consider that only 5-HT receptors are activated. Furthermore, 5-HT has a higher affinity (3- to 4-fold greater) for its own receptors than for $\alpha_{1/2}$ -adrenoceptors. This rules out or at least minimizes the role of α -adrenoceptors in our experimental condition. Furthermore, the antagonists were used at doses high enough to block their respective receptors in behavioral experiments (Condés-Lara et al., 2009; Lozano-Cuenca et al., 2005; Holmes et al., 2001).

Our results support the participation of the OTR and the 5-HT receptors in the OXT- and 5-HT-induced behavioral antinociception. In the case of 5-HT receptors, it is important to keep in mind that modulation of nociception at the spinal level by 5-HT receptors is complex, as they may either inhibit or facilitate nociceptive transmission. Previous studies in acute pain models have shown that i.t. 5-HT evoked antinociception (Bardin et al., 1997; Schmauss et al., 1983; Xu et al.,

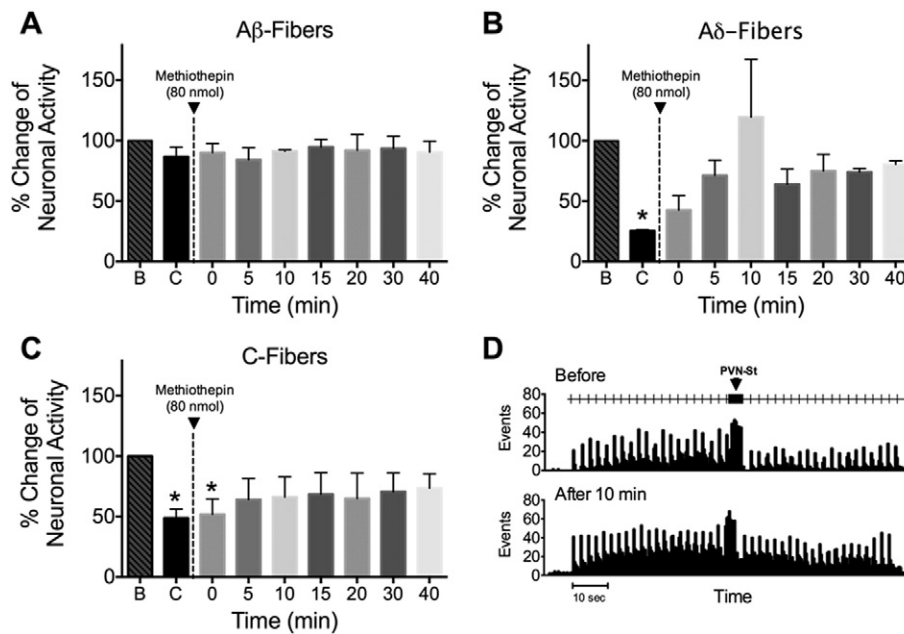


Fig. 6. Spinal methiothepin reduces the PVN-induced antinociception. Spinal methiothepin (80 nmol, $n = 3$) partially reverses the antinociceptive effect of PVN stimulation. Note that PVN stimulation or methiothepin administration (dotted line) does not have any effect on the activity of Aβ-fibers (panel A). On the other hand, the activity of Aδ- and C-fibers (panels B and C) was blocked by PVN stimulation, an effect reduced by spinal methiothepin. * $p < 0.05$ vs. B (basal activity). In panel D, the effect of methiothepin on the PVN-induced antinociception is depicted. Note that before methiothepin administration (upper trace) the PVN stimulation (PVN-St) induces a clear inhibition of the neuronal activity evoked by peripheral electrical stimulation; this PVN-induced inhibition is partially reverted after (10 min) methiothepin administration (lower trace).

1994; Yaksh and Wilson, 1979; Jeong et al., 2012) by activating the spinal 5-HT₁ (Godínez-Chaparro et al., 2013; Mjellem et al., 1992; Muñoz-Islas et al., 2014) and 5-HT_{5A} (Muñoz-Islas et al., 2014) receptors. In contrast, 5-HT_{2/3/4/6/7} receptors produce an increase of both acute and long-term nociceptive behavior (Bravo-Hernández et al., 2012; Cervantes-Durán et al., 2013; Godínez-Chaparro et al., 2011). Thus, we propose that the 5-HT receptors involved in the antinociception in our experiments could be associated with the activation of 5-HT₁ and 5-HT₅ receptors, both of which are coupled to signal transduction systems that participate in hyperpolarization by inhibiting adenylyl cyclase activity and/or by activating inwardly rectifying K⁺ channels (Nestler et al., 2009). Admittedly, further experiments are needed to determine the 5-HT receptor type(s) and subtype(s) involved.

4.4. Possible involvement of spinal 5-HT receptors in the inhibition by PVN electrical stimulation of the evoked nociceptive activity of WDR cells

As previously reported, electrical PVN stimulation selectively reduces the neuronal activity of Aδ- and C-fibers (Fig. 4G and H). In

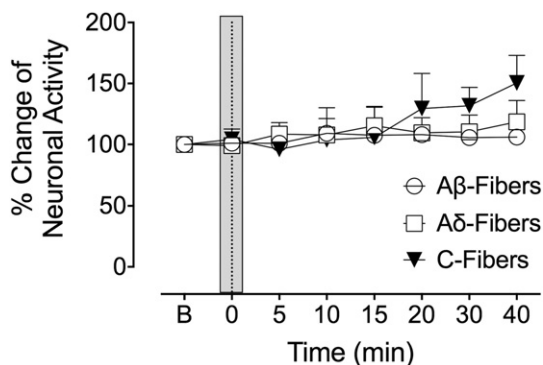


Fig. 7. Effects of methiothepin *per se* on the WDR neuronal activity. Spinal methiothepin (80 nmol, $n = 5$) by itself does not have any effect on the neuronal nociceptive evoked activity.

general, recordings of the WDR cells in healthy rats showed that the PVN-induced antinociception (particularly with the C-fibers) was enhanced after spinal 5-HT (Fig. 5). Certainly, spinal 5-HT alone or with the PVN stimuli has no substantial effect on the Aβ-fibers (Fig. 5A and B), but it does exert a biphasic effect on the activity of Aδ-fibers (*i.e.*, an increase of the neuronal activity in the first 10 min and a later inhibition) (Fig. 5C and D). The simplest interpretation of these findings could be related to the fact that different doses of serotonin could favor various specific responses (an increase or decrease of the neuronal activity) or possibly a dual effect (Oyama et al., 1996). Indeed, this monoamine has a high affinity for all 5-HT receptors (*i.e.*, 7 families {5-HT₁₋₇} and at least 14 receptor subtypes) and depending on the receptor activated, a depolarization or hyperpolarization could be induced (Hoyer et al., 2002). Accordingly, in freshly isolated rat dorsal root ganglion neurons, serotonin is able to enhance the cation current (*i.e.*, it increases the neuronal excitability) of medium-diameter DRG cells (*i.e.* Aδ-fiber type) but not C-fiber type (Cardenas et al., 1999). Moreover, in superficial dorsal horn neurons, serotonin can produce a transient suppression or facilitation of the excitatory postsynaptic currents (EPSCs) (Hori et al., 1996). Together, these data support our finding that serotonin has a biphasic effect on Aδ-fibers. In addition, although a biphasic effect is observed over time, when we express the values obtained as area under the curve, 5-HT alone has no effect on the activity of Aδ-fiber (Fig. 5G) in the first 10 min, but the activity of these fibers is diminished between 10 and 40 min. Surprisingly, the PVN-induced antinociception was blocked by 50 nmol 5-HT. We cannot explain this result, but we need to keep in mind that serotonin exerts complex effects, depending on the dose used, and the receptor(s) and neuronal population activated.

As previously demonstrated (Todorovic and Anderson, 1992), spinal 5-HT decreases the activity of C-fibers (particularly after 10 min; Fig. 5E, F and H). This inhibition was clearly enhanced by the PVN stimulation when the dose of 5-HT was 50 nmol (Fig. 5H). The enhancement of the inhibition could be associated with a synergistic effect or might reflect a direct interaction between the oxytocinergic and serotonergic systems. To partially answer this question, we performed additional experiments blocking the spinal 5-HT receptors and stimulating the PVN nuclei (Fig. 6).

It is interesting to note that the PVN-induced blockade of the activity of Aδ- and C-fibers was partially reverted with the topical spinal cord application of methiothepin, supporting the hypothesis that serotonergic mechanisms participate in the oxytocin-induced analgesia. At the dose used, methiothepin *per se* has no effect on the electrophysiological nociceptive activity (Fig. 7), supporting the use of this dose in the present experiments. These results do not exclude the possibility that the oxytocinergic and serotonergic systems act synergistically. In any case, our experiments support the hypothesis that there is an interaction between these two systems.

We have no direct evidence that similar results would be obtained in sensitized animals, since the reduction of the neuronal activity (in our case) was associated with inhibition of Aδ- and C-fibers but not of Aβ-fibers, which are the fibers activated during allodynia (Sandkühler, 2009). Although we cannot exclude the contribution of unmyelinated, low-threshold mechanoreceptors (*i.e.*, C-fibers) (Seal et al., 2009), these results suggest that an interaction between the oxytocinergic and serotonergic systems exists.

4.5. Conclusion

Although a synchronization of the PVN and RMg or other structures (*e.g.*, caudate nucleus) (Pan et al., 2016) may have a synergistic inhibitory role on nociception, an interaction at the spinal cord level between these two systems could exist. In fact, both oxytocin (Moreno-López et al., 2013) and 5-HT (Wang et al., 2009; Ahn and Basbaum, 2006; Zhang et al., 2002) receptors have been located in the superficial layers of the SDH, probably on the primary afferent fibers or GABAergic interneurons. This study provides only indirect evidence about the potential role of 5-HT in the PVN-induced descending inhibition, but the results are the first evidence *in vivo* suggesting that spinal antinociception produced by OXT could be partially mediated by the serotonergic system. In addition, this study raises future questions about the presence of both types of receptors (OTR and 5-HT receptors) on primary afferent fibers, interneurons or WDR cells on the SDH.

Conflict of interest

The authors declare no conflict of interest. The sponsors had no further role in (1) study design, (2) collection, analysis and interpretation of data, (3) the writing of the report and (4) the decision to submit the paper for publication.

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